DATA SCIENCE MINOR PROJECT REPORT

A Project report

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Customer Transaction Trends & Payment Analysis using Power BI (2022–2025)

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Discipline of CSE/IT

Lovely School of Computing Science

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DECLARATION

I, **Rishav Raj**, student of **B.Tech Computer Science and Engineering** under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 19/04/2026
Registration No: 12525964
Signature of the Student: Rishav Raj

CERTIFICATE

This is to certify that **Rishav Raj** bearing Registration no. **12525964** has completed the **INT-387** project titled, **"Customer Transaction Trends & Payment Analysis using Power BI (2022–2025)"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort, and study.

Signature and Name of the Supervisor: _____

Designation: _____

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Abstract

This project focuses on analysing customer transaction data to identify payment trends and behavioural patterns using Power BI. The dataset spans from 2022 to 2025 and includes various transaction modes such as credit cards and debit cards. The objective is to transform raw data into meaningful insights through data cleaning, visualization, and dashboard creation. The results highlight shifts in payment preferences and provide actionable insights for decision-making.

Keywords

Power BI, Data Visualization, Customer Transactions, Payment Trends, Dashboard, Data Analytics

1. INTRODUCTON TO THE PROJECT

With the increasing use of digital payments, analysing transaction data has become essential for understanding customer behaviour and financial patterns. In recent years, there has been a noticeable shift from cash-based transactions to digital modes such as credit cards, debit cards, and other electronic payment systems. This transition has led to the generation of large volumes of transactional data, which, when properly analysed, can reveal valuable insights about customer preferences, spending habits, and usage trends. Understanding these patterns helps organizations identify how customers interact with different payment methods and how their behaviour evolves over time.

This project aims to study transaction trends and compare different payment methods using interactive dashboards. By focusing on data from recent years, the project highlights variations in credit card and debit card usage, identifies key trends, and provides a clearer picture of customer financial behaviour. Power BI is used as the primary tool for visualization, enabling the transformation of raw data into meaningful charts, graphs, and key performance indicators. The use of interactive features such as filters and slicers further enhances data exploration, making it easier to derive insights and support effective decision-making.

2. Objectives

- To analyse customer transaction trends from 2022–2025
- To compare credit card and debit card usage
- To visualize key insights using dashboards
- To identify patterns in payment behaviour
- To enhance decision-making using data insights

3. Methodology

3.1 Data Collection

The dataset was collected from reliable sources and contains transaction details.

3.2 Data Cleaning

- Removed missing values
- Standardized formats
- Filtered relevant columns

3.3 Data Transformation

- Created calculated columns

3.4 Dashboard Creation

Interactive dashboards were created using charts, filters, and KPIs.

CREDIT CARD DASHBOARD

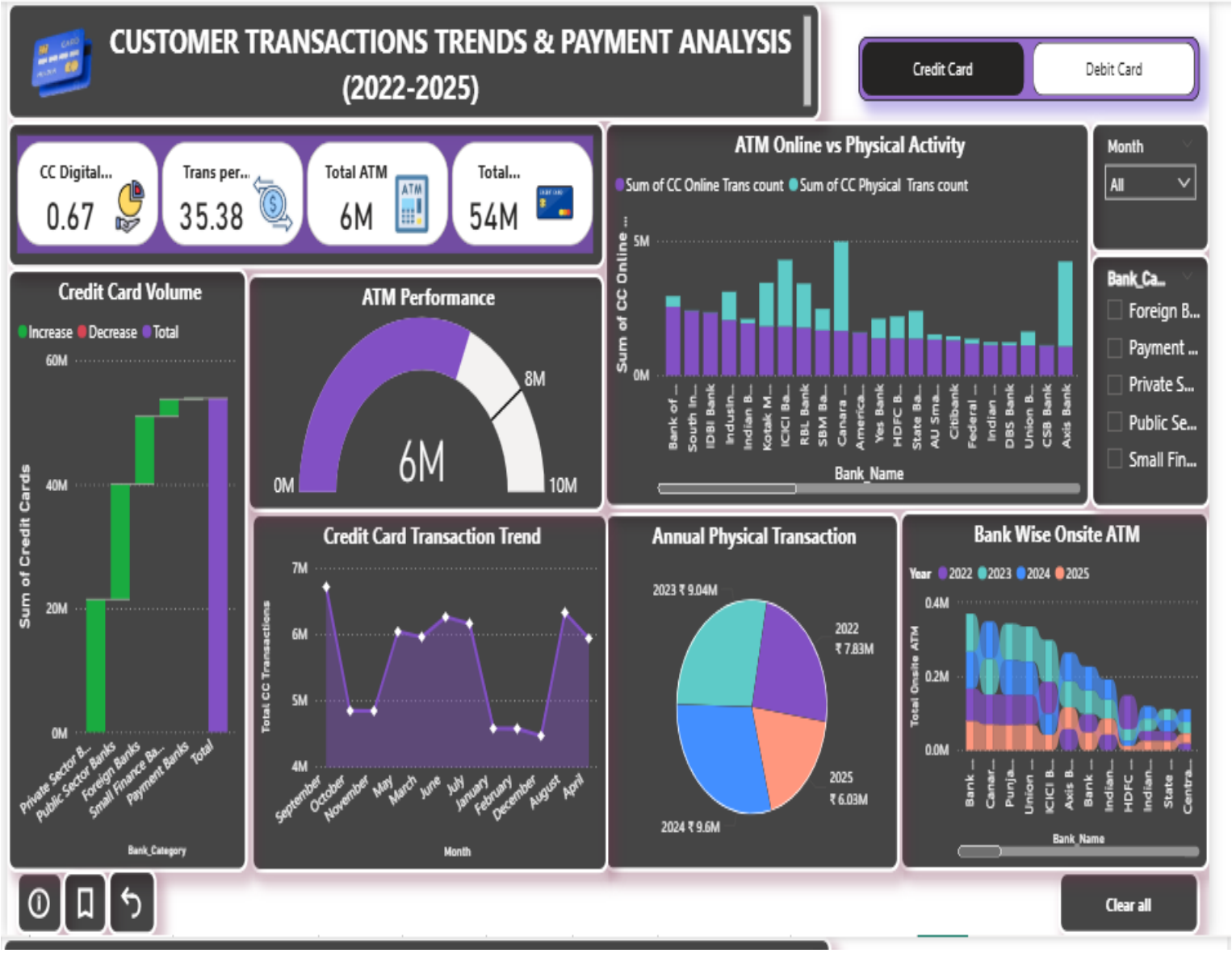


Fig.01

DEBIT CARD DASHBOARD

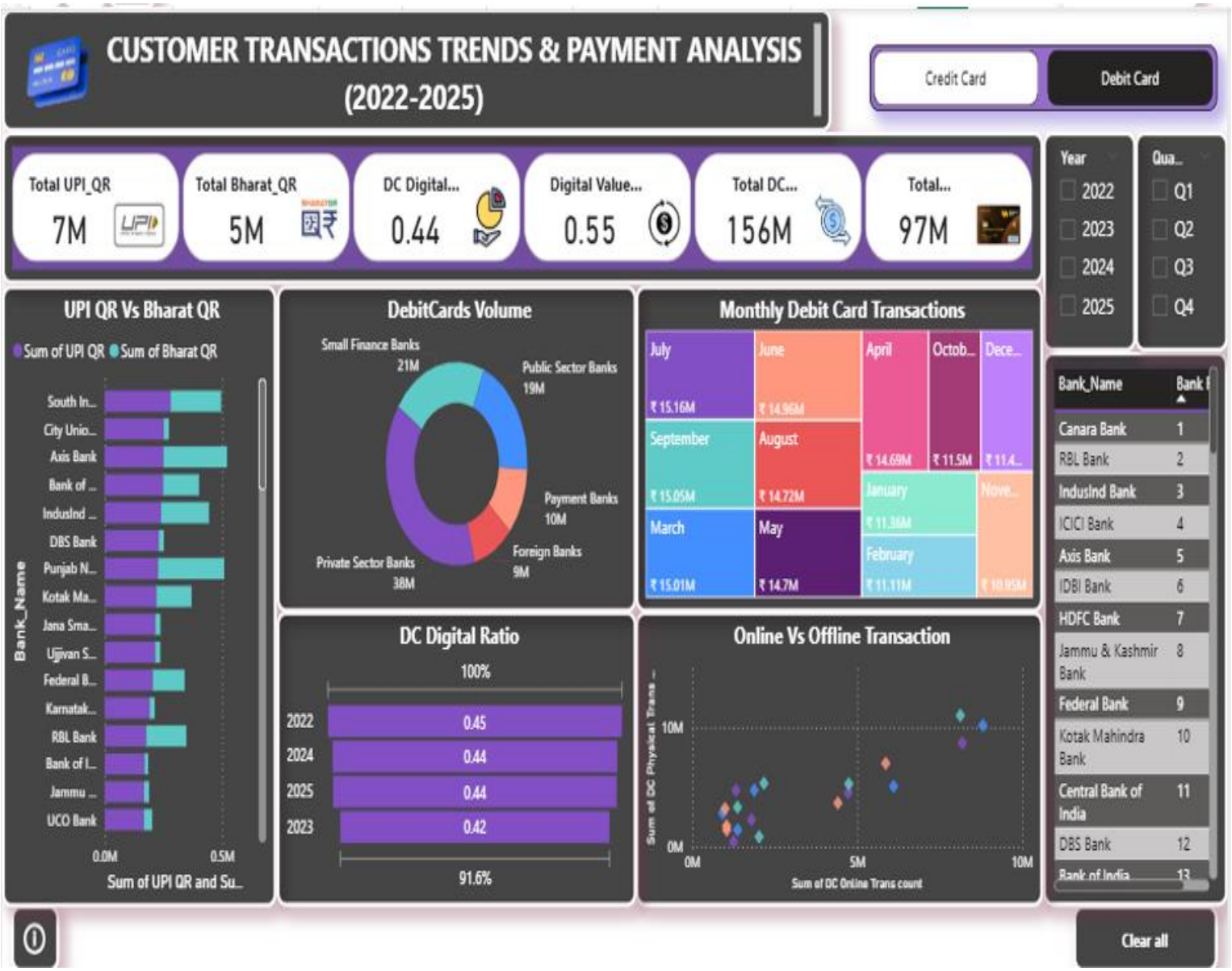


Fig.02

4. Tools & Technologies Used

This project utilizes a combination of tools and technologies to perform data analysis, cleaning, and visualization effectively. Each tool plays a significant role in transforming raw transaction data into meaningful insights.

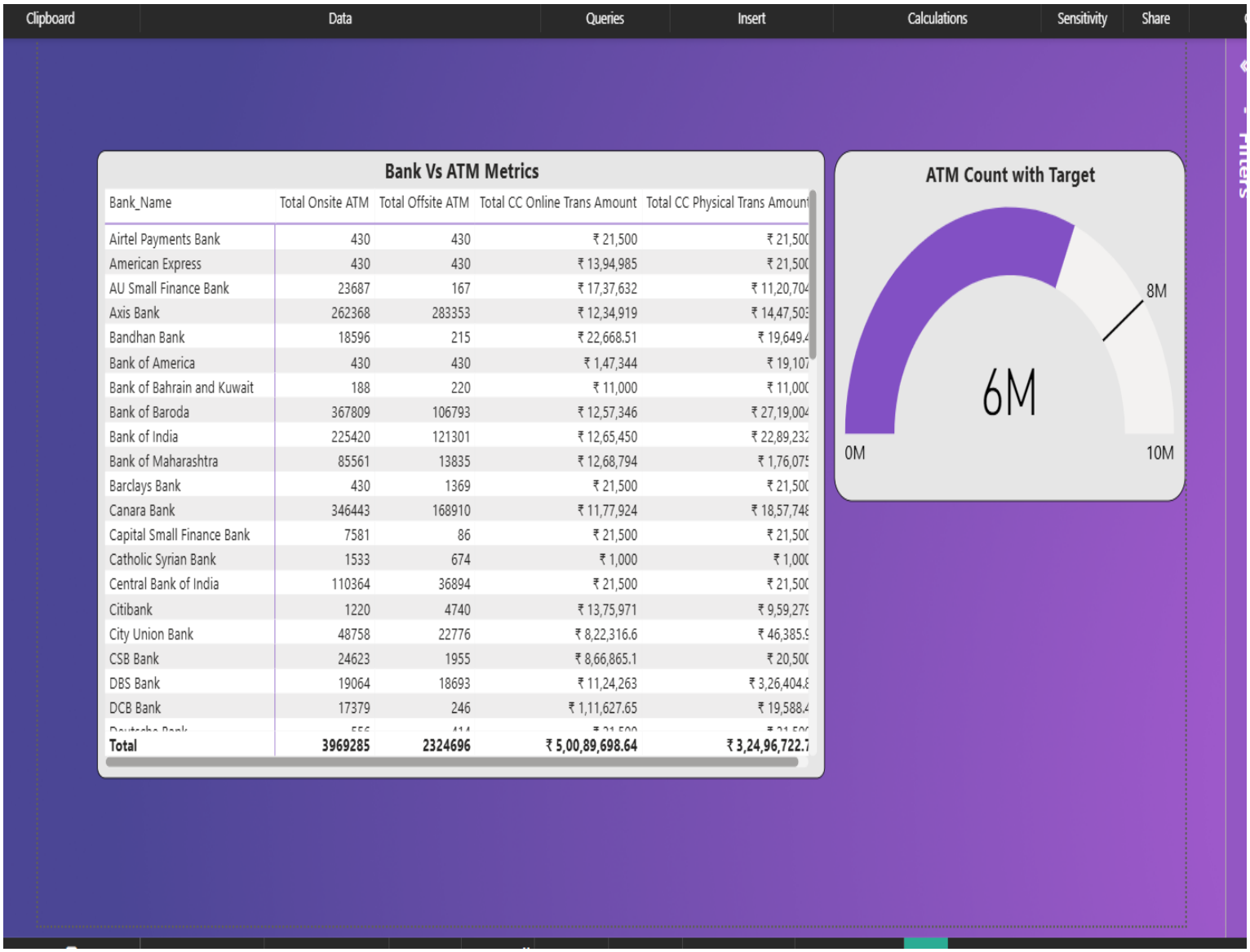
Power BI:
Power BI is the primary tool used for data visualization and dashboard creation in this project. It is a powerful business intelligence platform that allows users to connect to various data sources, transform raw data, and create interactive visual reports. In this project, Power BI was used to design dashboards that represent transaction trends, compare credit card and debit card usage, and display key performance indicators (KPIs). Features such as filters, slicers, and drill-down capabilities were used to enhance user interaction and enable dynamic data exploration. Power BI also provides various visualization options like bar charts, line graphs, pie charts, and cards, which help in presenting data in a clear and visually appealing manner.

Microsoft Excel:
Microsoft Excel was used as the initial platform for handling and preparing the dataset. It helped in organizing the data in a structured format and performing basic preprocessing tasks such as sorting, filtering, and removing duplicate or unnecessary entries. Excel functions were also used to clean and validate the data before importing it into Power BI. Its tabular format made it easier to understand the dataset and perform initial analysis.

Data Cleaning Techniques:
Data cleaning is an essential step in any data analysis project. In this project, various data cleaning techniques were applied to improve the quality of the dataset. These include handling missing values, removing duplicate records, correcting inconsistent data formats, and selecting relevant columns for analysis. Proper data cleaning ensures accuracy and reliability of the results, which is crucial for generating meaningful insights.

Visualization Techniques and Tools:
Different visualization techniques were used to represent the data effectively.

Charts such as bar graphs, line charts, and pie charts were used to show comparisons, trends, and proportions in the dataset. KPI cards were used to highlight important metrics such as total transactions and average values. The use of appropriate visualization techniques helps in simplifying complex data and makes it easier for users to interpret the results. Interactive elements like filters and slicers further improve usability by allowing users to customize the view according to their needs.



Bank_Name	Total Onsite ATM	Total Offsite ATM	Total CC Online Trans Amount	Total CC Physical Trans Amount
Airtel Payments Bank	430	430	₹ 21,500	₹ 21,500
American Express	430	430	₹ 13,94,985	₹ 21,500
AU Small Finance Bank	23687	167	₹ 17,37,632	₹ 11,20,704
Axis Bank	262368	283353	₹ 12,34,919	₹ 14,47,503
Bandhan Bank	18596	215	₹ 22,668.51	₹ 19,649.4
Bank of America	430	430	₹ 1,47,344	₹ 19,107
Bank of Bahrain and Kuwait	188	220	₹ 11,000	₹ 11,000
Bank of Baroda	367809	106793	₹ 12,57,346	₹ 27,19,004
Bank of India	225420	121301	₹ 12,65,450	₹ 22,89,232
Bank of Maharashtra	85561	13835	₹ 12,68,794	₹ 1,76,075
Barclays Bank	430	1369	₹ 21,500	₹ 21,500
Canara Bank	346443	168910	₹ 11,77,924	₹ 18,57,748
Capital Small Finance Bank	7581	86	₹ 21,500	₹ 21,500
Catholic Syrian Bank	1533	674	₹ 1,000	₹ 1,000
Central Bank of India	110364	36894	₹ 21,500	₹ 21,500
Citibank	1220	4740	₹ 13,75,971	₹ 9,59,279
City Union Bank	48758	22776	₹ 8,22,316.6	₹ 46,385.9
CSB Bank	24623	1955	₹ 8,66,865.1	₹ 20,500
DBS Bank	19064	18693	₹ 11,24,263	₹ 3,26,404.8
DCB Bank	17379	246	₹ 1,11,627.65	₹ 19,588.4
Total	**3969285**	**2324696**	**₹ 5,00,89,698.64**	**₹ 3,24,96,722.7**

Fig.03

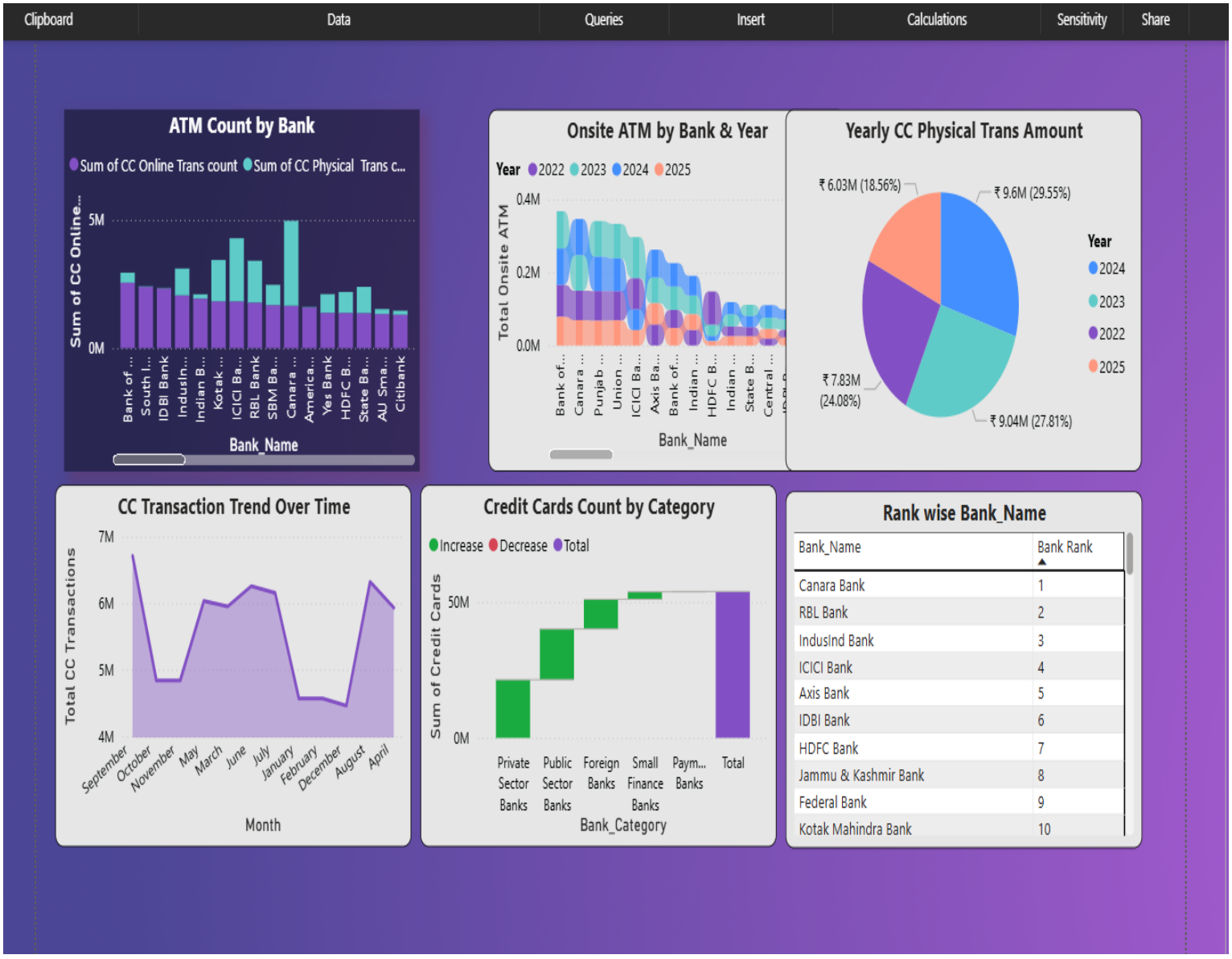

Clipboard
Data
Queries
Insert
Calculations
Sensitivity
Share
ATM Count by Bank
Sum of CC Online Trans count
Sum of CC Physical Trans c...
Sum of CC Online...
5M
0M
Bank of ...
South I...
IDBI Bank
IndusIn...
Indian B...
Kotak ...
ICICI Ba...
RBL Bank
SBM Ba...
Canara ...
America...
Yes Bank
HDFC B...
State Ba...
AU Sma...
Citibank
Bank_Name
Onsite ATM by Bank & Year
Year
2022
2023
2024
2025
Total Onsite ATM
0.4M
0.2M
0.0M
Bank of...
Canara ...
Punjab ...
Union ...
ICICI Ba...
Axis Ba...
Bank of...
Indian ...
HDFC B...
Indian ...
State B...
Central ...
Bank_Name
Yearly CC Physical Trans Amount
₹ 6.03M (18.56%)
₹ 9.6M (29.55%)
₹ 7.83M (24.08%)
₹ 9.04M (27.81%)
Year
2024
2023
2022
2025
CC Transaction Trend Over Time
Total CC Transactions
7M
6M
5M
4M
September
October
November
May
March
June
July
January
February
December
August
April
Month
Credit Cards Count by Category
Increase
Decrease
Total
Sum of Credit Cards
50M
0M
Private Sector Banks
Public Sector Banks
Foreign Banks
Small Finance Banks
Paym... Banks
Total
Bank_Category
Rank wise Bank_Name
Bank_Name
Bank Rank
Canara Bank 1
RBL Bank 2
IndusInd Bank 3
ICICI Bank 4
Axis Bank 5
IDBI Bank 6
HDFC Bank 7
Jammu & Kashmir Bank 8
Federal Bank 9
Kotak Mahindra Bank 10

Fig.04

5. Results & Analysis

The developed dashboard provides a comprehensive view of customer transaction behaviour and reveals several meaningful insights through visual analysis. By representing the data using interactive charts and key performance indicators, it becomes easier to identify patterns, trends, and variations in payment activities over the given time period.

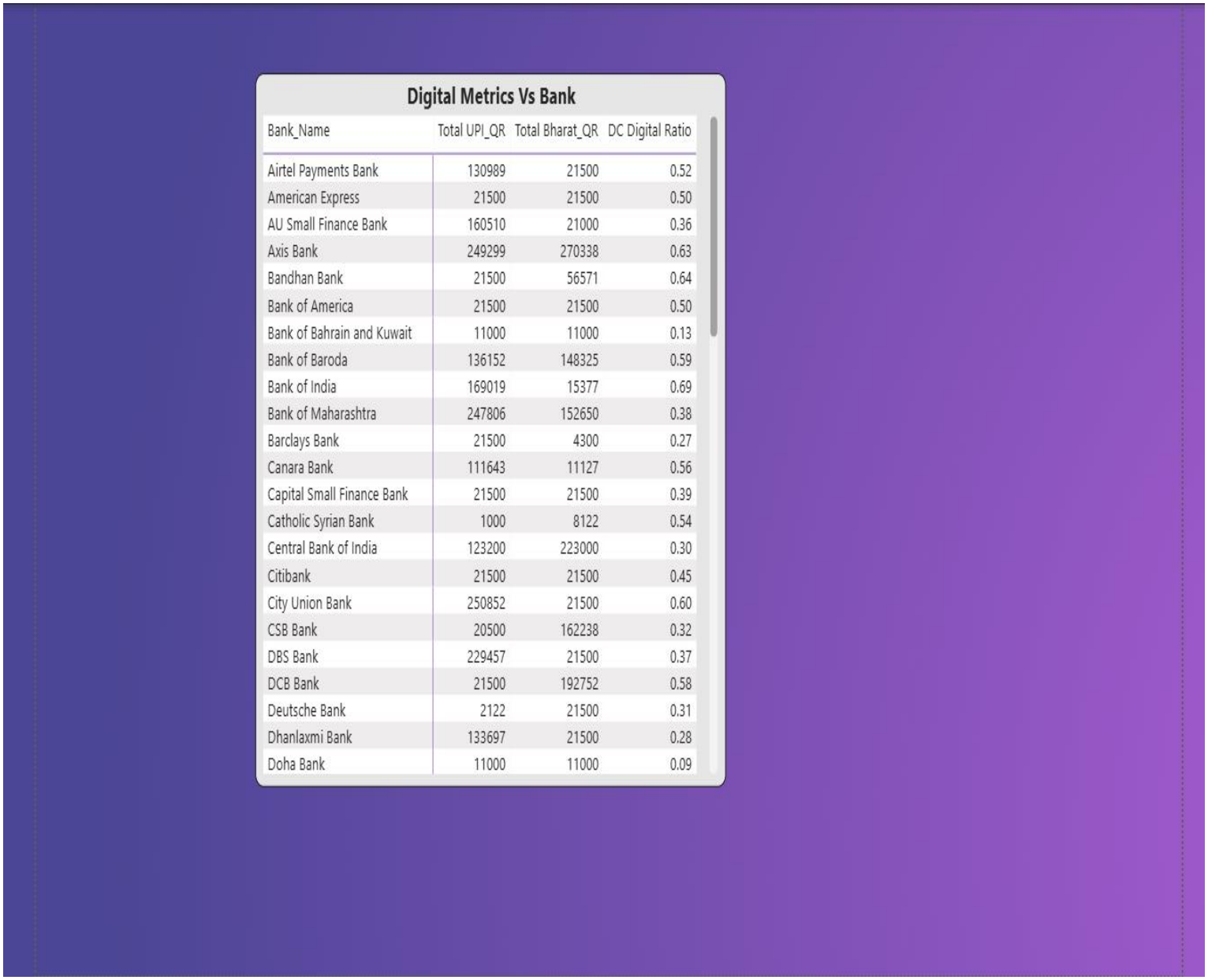
One of the major observations from the dashboard is the **increasing trend in digital payments** over the years from 2022 to 2025. The visualizations clearly indicate a steady growth in the number and value of transactions, which reflects the rising adoption of digital payment methods. This trend highlights the growing trust of customers in digital financial systems and the shift towards a cashless economy.

The dashboard also presents a detailed **comparison between credit card and debit card usage**. It can be observed that both payment methods have distinct usage patterns. In certain periods, credit card transactions show higher values, possibly due to larger purchases or offers associated with credit usage, while debit card transactions may reflect more frequent but smaller-value transactions. This comparison helps in understanding customer preferences and financial behaviour.

Another important insight is related to **monthly and yearly transaction patterns**. The data shows fluctuations in transaction volumes across different months, indicating seasonal trends or peak activity periods. For example, certain months may experience higher transaction volumes due to festivals, sales, or financial cycles. Yearly comparisons further help in identifying overall growth trends and long-term changes in customer behaviour.

Finally, the dashboard provides valuable **customer behaviour insights** by highlighting how users interact with different payment methods over time. It helps in identifying patterns such as preferred payment modes, frequency of transactions, and changes in spending habits. These insights are useful for businesses and financial institutions to improve their services, target customers more effectively, and make data-driven decisions.

Overall, the results demonstrate how data visualization using Power BI can transform raw transactional data into meaningful and actionable insights.



Digital Metrics Vs Bank

Bank_Name	Total UPI_QR	Total Bharat_QR	DC Digital Ratio
Airtel Payments Bank	130989	21500	0.52
American Express	21500	21500	0.50
AU Small Finance Bank	160510	21000	0.36
Axis Bank	249299	270338	0.63
Bandhan Bank	21500	56571	0.64
Bank of America	21500	21500	0.50
Bank of Bahrain and Kuwait	11000	11000	0.13
Bank of Baroda	136152	148325	0.59
Bank of India	169019	15377	0.69
Bank of Maharashtra	247806	152650	0.38
Barclays Bank	21500	4300	0.27
Canara Bank	111643	11127	0.56
Capital Small Finance Bank	21500	21500	0.39
Catholic Syrian Bank	1000	8122	0.54
Central Bank of India	123200	223000	0.30
Citibank	21500	21500	0.45
City Union Bank	250852	21500	0.60
CSB Bank	20500	162238	0.32
DBS Bank	229457	21500	0.37
DCB Bank	21500	192752	0.58
Deutsche Bank	2122	21500	0.31
Dhanlaxmi Bank	133697	21500	0.28
Doha Bank	11000	11000	0.09

Fig.05


UPI QR Vs Bharat QR by Bank
Sum of UPI QR
Sum of Bharat QR
South In...
City Uni...
Axis Bank
Bank of ...
IndusInd...
DBS Bank
Punjab ...
Kotak M...
Jana Sm...
Ujjivan S...
Federal ...
Karnatak...
RBL Bank
Bank of I...
Jammu ...
UCO Bank
HDFC Ba...
AU Smal...
Indian O...
State Ba...
Bank_Name
0.0M
0.5M
Sum of UPI QR and S...
Category Wise DebitCards
9M
(9.42%)
38M
(38.72%)
10M
(10.48%)
19M
(19....)
21M (21.76%)
Bank_Category
Private Secto...
Small Financ...
Public Sector...
Payment Ban...
Foreign Banks
Total DC Trans Amount by Month
July
₹ 15.16M
June
₹ 14.96M
October
₹ 11.5M
December
₹ 11.46M
August
₹ 14.72M
September
₹ 15.05M
May
₹ 14.7M
January
₹ 11.36M
February
₹ 11.11M
March
₹ 15.01M
April
₹ 14.69M
November
₹ 10.95M
Online Vs Physical Trans count by Category and Year
Year
2022
2023
2024
2025
Sum of DC Physical Trans count
10M
5M
0M
0M
5M
10M
Sum of DC Online Trans count
DC Digital Ratio by Year
100%
2022
0.45
2024
0.44
2025
0.44
2023
0.42
91.6%

Fig.06

6. Discussion

The analysis of the dashboard clearly indicates a gradual shift of customers towards digital payment methods over the period from 2022 to 2025. This trend reflects the growing adoption of technology, increased accessibility of online payment systems, and greater trust in digital financial services. The consistent rise in transaction volume suggests that customers are becoming more comfortable with cashless transactions, which aligns with the broader movement towards a digital economy. This shift is also supported by factors such as convenience, faster processing, and the availability of various digital payment options.

A deeper examination of the data highlights noticeable differences in credit card and debit card usage patterns, indicating varied financial behaviour among customers. Credit card transactions often show higher values, which may be associated with larger purchases, credit-based spending, or promotional offers such as cashback and rewards. In contrast, debit card usage tends to reflect more controlled spending, as it is directly linked to the user's bank balance. This difference suggests that customers use credit cards for flexibility and benefits, while debit cards are preferred for routine and budget-conscious transactions.

Furthermore, the variation in transaction patterns across different time periods suggests that external factors such as seasonal demand, economic conditions, and promotional events may influence customer behaviour. Peaks in transaction activity during certain months indicate increased spending during festive seasons or special sales periods. These insights emphasize the importance of understanding customer behaviour not only in terms of payment methods but also in relation to time-based trends.

Overall, the discussion highlights that customer payment behaviour is dynamic and influenced by multiple factors, including convenience, financial flexibility, and external conditions. The use of data visualization through dashboards provides a clear and effective way to interpret these patterns. Such insights can be valuable for businesses and financial institutions in designing better strategies, improving customer experience, and adapting to the evolving digital payment landscape.

7. Conclusion

This project successfully demonstrates the importance of data visualization in transforming raw transactional data into meaningful and actionable insights. By analysing customer transaction data from 2022 to 2025, clear patterns and trends in digital payment behaviour were identified. The use of interactive dashboards made it easier to interpret complex data, compare different payment methods, and observe changes in transaction trends over time. This highlights how visual tools can simplify data analysis and support better understanding of customer behaviour.

The implementation of dashboards using Power BI proved to be highly effective in presenting the data in a clear and structured manner. Features such as charts, filters, and key performance indicators helped in delivering a comprehensive view of the analysis. Overall, the project emphasizes the role of business intelligence tools in enabling data-driven decision-making and provides valuable insights that can be used by businesses to enhance their strategies and improve customer experience.

8. Future Scope

The current project focuses on analysing historical transaction data; however, there are several opportunities to enhance and expand its capabilities in the future. One of the major improvements can be the **integration of real-time data**. By connecting the dashboard to live data sources, such as banking APIs or streaming datasets, the system can provide up-to-date insights and enable real-time monitoring of transactions. This would significantly improve decision-making by allowing organizations to respond quickly to changing trends and customer behaviour.

Another important area for future development is the use of **machine learning techniques for predictive analysis**. By applying algorithms such as regression, classification, or time-series forecasting, the system can predict future transaction trends, customer preferences, and spending patterns. This would add a predictive layer to the project, moving beyond descriptive analytics to more advanced, data-driven forecasting. Additionally, expanding the dataset to

include more variables—such as customer demographics, geographic data, and additional payment methods—can provide deeper insights and a more comprehensive understanding of customer behaviour.

9. References

[1] Microsoft Corporation, “Microsoft Power BI Documentation.” [Online]. Available: https://learn.microsoft.com/power-bi

[2] Microsoft Corporation, “Microsoft Excel Documentation.” [Online]. Available: https://support.microsoft.com/excel

[3] F. Provost and T. Fawcett, *Data Science for Business*, Sebastopol, CA, USA: O’Reilly Media, 2013.

[4] W. McKinney, *Python for Data Analysis*, 2nd ed., Sebastopol, CA, USA: O’Reilly Media, 2017.

[5] National Payments Corporation of India (NPCI), “Retail Payments Statistics and Transaction Data.” [Online]. Available: https://www.npci.org.in/statistics

[6] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Burlington, MA, USA: Morgan Kaufmann, 2011.

[7] IEEE Xplore, “Research Papers on Data Analytics and Visualization.” [Online]. Available: https://ieeexplore.ieee.org

[8] Google Scholar, “Academic Research on Business Intelligence and Data Visualization.” [Online]. Available: https://scholar.google.com

9. PROJECT LINK

- **LinkedIn Link:** https://www.linkedin.com/in/rishavraj0001/
- **GitHub Link:** https://github.com/RishavRaj110/Customer-Transaction-Trends-Payment-Analysis-.git